



Tribhuvan University
Institution of Science and Technology
A Final Year Project Report
On
AGROTREE: A DECISION TREE BASED CROP
RECOMMENDATION SYSTEM

Under the Supervision of

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Submitted To:

**DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION
TECHNOLOGY**

ORCHID INTERNATIONAL COLLEGE

**In partial fulfillment for the Bachelor's Degree in Computer Science
and Information Technology**

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SUPERVISOR'S RECOMMENDATION

I hereby recommend that this project prepared under my supervision by Dikshan Kc (79010772), Chetan Bhattarai(79010771), Prajwal Pathak(79010785) entitled **“AgroTree: A Decision Tree Based Crop Recommendation System”** in partial fulfilment of the requirements for the degree of B.Sc in Computer Science and Information Technology is recommended for the final evaluation.

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CERTIFICATE OF APPROVAL

This is to certify that the project entitled “**AgroTree: A Decision Tree Based Crop Recommendation System**” submitted by Dikshan KC (79010772), Chetan Bhattarai (79010771), and Prajwal Pathak (79010785) in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Information Technology (B.Sc. CSIT) has been successfully completed at Orchid International College, Tribhuvan University.

We hereby declare that the project work carried out by the students is original and has been conducted in accordance with the academic standards and guidelines prescribed by the institution. The project demonstrates satisfactory knowledge, practical implementation, research capability, and technical skills in the field of computer science and information technology.

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ABSTRACT

Efficient crop selection is essential for improving agricultural productivity and promoting sustainable farming practices. This system presents **AgroTree: A Decision Tree Based Crop Recommendation System**, which predicts the most suitable crop for cultivation based on soil and climatic parameters such as Nitrogen (N), Phosphorus (P), Potassium (K), pH value, temperature, humidity, and rainfall. The dataset used for training and testing the model was collected from Kaggle and contains historical agricultural data related to crop suitability. The system uses a Decision Tree algorithm based on the CART (Classification and Regression Tree) approach, implemented from scratch without relying entirely on pre-built machine learning models. The algorithm utilizes the Gini Index as the splitting criterion to determine the best feature splits during tree construction. This implementation helps in understanding fundamental machine learning concepts such as node splitting, impurity calculation, recursive partitioning, and tree generation. The frontend of the system is developed using Streamlit, while Python is used for backend processing and model implementation. Supabase is integrated as the database for user authentication and prediction history management. The system provides accurate and interpretable crop recommendations, helping farmers make informed decisions, reduce crop failure risks, and improve resource utilization.

Keywords:

Crop Recommendation System, Decision Tree, CART, Gini Index, Machine Learning, Streamlit, Supabase, Precision Agriculture.

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LIST OF ABBREVIATIONS

AgroTree - A Decision Tree Based Crop Recommendation System

API - Application Programming Interface

CART - Classification and Regression Tree

CSV - Comma Separated Values

DT - Decision Tree

ML - Machine Learning

NPK - Nitrogen, Phosphorus, Potassium

UML - Unified Modeling Language

SDLC - Software Development Life Cycle

pH - Potential of Hydrogen

UI - User Interface

REST- Representational State Transfer

CHAPTER 1-INTRODUCTION

1.1 Introduction

Agriculture is a major source of livelihood in developing countries like Nepal, where selecting the right crop is essential for improving productivity and ensuring sustainable farming practices. Traditional farming methods often rely on farmers' experience and historical knowledge, which may not always provide accurate results due to changing environmental and climatic conditions. To address this issue, data-driven approaches using Machine Learning can support farmers in making better agricultural decisions. This system, “**AgroTree: A Decision Tree Based Crop Recommendation System,**” uses soil nutrients such as Nitrogen (N), Phosphorus (P), Potassium (K), along with pH value, temperature, rainfall, and humidity to recommend the most suitable crop for cultivation. The system is developed using the Decision Tree algorithm based on the CART (Classification and Regression Tree) approach. The algorithm uses the Gini Index as the splitting criterion for generating accurate predictions. The system is implemented as a web-based application using Streamlit, providing real-time crop recommendations through a simple and user-friendly interface. In addition, Supabase is integrated for user authentication and prediction history management.

Table 1.1: Agricultural Input Parameters Used in Crop Prediction

Parameters	Valid Range (Hard)	Unusual Low Trigger	Unusual High Trigger	Near- boundary Margin (8%)
Nitrogen (N)	0 – 155 kg/ha	< 5	> 125	± 12.3
Phosphorus (P)	4 – 170 kg/ha	< 12	> 138	±10.8
Potassium (K)	4 – 235 kg/ha	< 8	> 205	± 18.4
pH Value	2.9 – 11	< 5.5	> 7.5	± 0.48
Temperature	8°C – 52°C	< 17	> 36	± 2.96
Humidity	13% – 100%	< 14	> 97	± 6.88
Rainfall	17– 845 mm	< 22	> 250	± 66.4

1.2 Problem Statement

Agricultural productivity depends on selecting suitable crops based on soil and environmental conditions. However, many farmers still rely on traditional knowledge, which may not be accurate due to changing climate and soil conditions.

Incorrect crop selection can result in low yield, higher costs, and financial losses. Existing systems are often complex or not user-friendly, making them less accessible to farmers. Therefore, there is a need for a simple and efficient system that provides accurate crop recommendations. This project addresses the problem by developing a Decision Tree-based crop recommendation system that supports better decision-making and improves agricultural productivity.

1.3 Objectives

- To develop a Decision Tree-based crop recommendation system that suggests the most suitable crop for given soil and climatic conditions
- To assist farmers in making informed, data-driven decisions for crop selection
- To design a simple, interactive, and user-friendly web interface using Streamlit.

1.4 Scope and Limitation

Scope:

The system focuses on recommending crops based on key input parameters such as Nitrogen (N), Phosphorus (P), Potassium (K), pH level, temperature, rainfall, and humidity. It is designed as a web-based application, making it accessible through standard web browsers. The system supports user interaction through a simple interface, allowing easy input and output visualization. The model is trained using a publicly available dataset, making it adaptable and extendable. The system can be further enhanced by integrating additional features such as weather APIs, mobile application support, and advanced machine learning models.

Limitations:

The accuracy of the system depends on the quality and scope of the dataset used for training. The system does not currently incorporate real-time weather data, which may affect prediction accuracy under dynamic conditions. It focuses only on crop recommendation and does not include fertilizer or irrigation recommendations. The system is dependent on user input, and incorrect input values may lead to inaccurate results.

1.5 Development Methodology

The development of the AgroTree system follows the **Agile Software Development Methodology**, which is an iterative and incremental approach used for developing flexible and efficient software systems. Agile methodology allows continuous improvement, regular testing, and gradual enhancement of the system throughout the development lifecycle. The system was developed in multiple phases, where each phase focused on a specific component of the crop recommendation system.

The major stages involved in the development process are described below.

1.5.1 Requirement Analysis

During this stage, the system requirements and user needs were identified and analyzed. The agricultural problems related to improper crop selection were studied to understand the need for an intelligent crop recommendation system. Functional requirements such as user authentication, crop prediction, and prediction history management were identified along with non-functional requirements including usability, performance, and security. The overall scope, objectives, and limitations of the system were also defined in this phase.

Input

- User requirements
- Agricultural problem domain
- Existing crop recommendation studies

Output / Artifacts

- Requirement Specification Document
- Functional Requirement List
- Non-Functional Requirement List
- Initial system scope definition

1.5.2 Data Collection and Preprocessing

In this stage, the agricultural dataset was collected from Kaggle. The dataset was analyzed to identify important features such as Nitrogen (N), Phosphorus (P), Potassium (K), pH value, temperature, humidity, and rainfall. Data preprocessing activities including data cleaning, handling missing values, formatting, and feature selection were performed to improve data quality. The dataset was then divided into training and testing datasets for machine learning model development and evaluation.

Input

Raw agricultural dataset

Output / Artifacts

Cleaned dataset

Processed training dataset

Testing dataset

Feature matrix and target labels

1.5.3 Mathematical Model

In this phase, the Decision Tree algorithm based on the CART methodology was implemented for crop prediction. The Gini Index was used as the splitting criterion to determine the best feature splits during tree construction. The model was trained using the prepared dataset, and recursive partitioning was performed to create decision nodes and leaf nodes. The trained model was evaluated using testing data to measure its prediction performance and accuracy.

Input Variables

Let:

$$X = \{x_1, x_2, x_3, \dots, x_n\}$$

Where:

x_1 = Nitrogen (N)

x_2 = Phosphorus (P)

x_3 = Potassium (K)

x_4 = pH value

x_5 = Temperature

x_6 = Humidity

x_7 = Rainfall

The output variable is:

$$Y = \text{Recommended Crop}$$

Gini Index Formula

The CART algorithm uses the Gini Index to measure node impurity.

$$Gini(D) = 1 - \sum_{i=1}^c (p_i)^2$$

Where:

D = Dataset at a node

c = Number of classes

p_i = Probability of class i

The algorithm selects the feature split with the minimum Gini impurity.

Decision Tree Function

The prediction function of the model can be represented as:

$$f(X) \rightarrow Y$$

Where:

X = Input feature vector

Y = Predicted crop label

Model Workflow

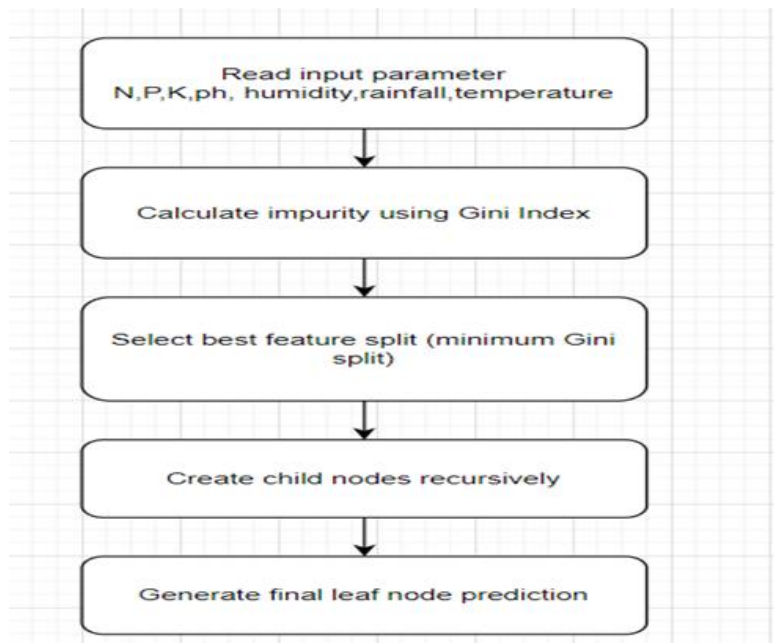


Figure 1.1 Model Workflow

Input

Preprocessed dataset

Output / Artifacts

- Trained Decision Tree model
- Model performance metrics
- Serialized .pkl model file

1.5.4 System Design

During the system design phase, the overall architecture of the AgroTree system was designed. The frontend interface, backend processing modules, and database structure were planned and organized. UML diagrams such as Use Case Diagram, Activity Diagram, Sequence Diagram, and ER Diagram were prepared to represent system workflows and interactions. User interface layouts were also designed to ensure simplicity and usability for end users.

Input

- Requirement specifications
- Processed dataset
- System objectives

Output / Artifacts

- Use Case Diagram
- Activity Diagram
- Sequence Diagram
- System Architecture Design

1.5.5 Implementation

In this stage, the actual development of the system was carried out using Python and Streamlit. The frontend interface was developed to allow users to input soil and climatic parameters and receive crop recommendations. The trained Decision Tree model was integrated into the application for real-time prediction. Supabase was integrated to implement user authentication and prediction history storage functionalities. Different modules of the system were developed and connected to ensure smooth system operation.

Input

- System design documents
- Trained machine learning model

Output / Artifacts

- Functional web application
- Integrated crop recommendation system
- User authentication system

1.5.6 Testing and Evaluation

Testing and evaluation were performed to verify the functionality and reliability of the system. Unit testing was conducted on individual modules such as input validation,

authentication, and prediction modules. System testing was performed to evaluate the complete workflow of the application. The machine learning model was evaluated using performance metrics such as accuracy, precision, recall, F1-score, and confusion matrix to analyze the effectiveness of crop prediction.

Input

- Trained model
- Testing dataset
- User test scenarios

Output / Artifacts

- Test case reports
- Accuracy results
- Performance evaluation report
- Final validated system

1.5.7 Deployment and Maintenance

In the final stage, the AgroTree system was deployed as a web-based application. The Streamlit application and Supabase database were configured to work together in an integrated environment. After deployment, system monitoring, bug fixing, and performance improvements were carried out to maintain system stability and functionality. Future updates and enhancements were also considered during this stage.

Input

- Final tested system

Output / Artifacts

- Deployed AgroTree system
- Operational crop recommendation platform
- Updated system versions

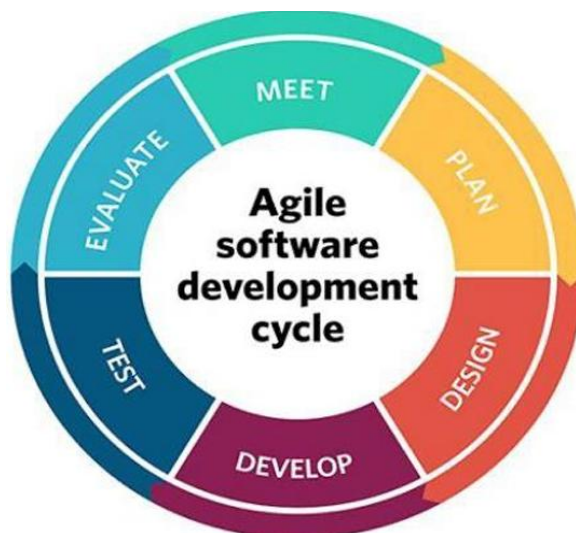


Figure 1.2:Agile Model

1.6 Report Organization

This report is organized into six chapters, each focusing on different aspects of the AgroTree Crop Recommendation System.

Chapter 1: Introduction

This chapter introduces the project background, problem statement, objectives, scope and limitations, development methodology, and overall organization of the report.

Chapter 2: Background Study and Literature Review

This chapter discusses the fundamental concepts related to crop recommendation systems, machine learning, Decision Tree algorithms, and reviews related research works and existing systems.

Chapter 3: System Analysis

This chapter presents the requirement analysis, feasibility analysis, and system modeling techniques including UML diagrams and system workflows used in the AgroTree system.

Chapter 4: System Design

This chapter describes the detailed system design, architecture, UML design diagrams, database structure, interface design, and algorithm details of the proposed system.

Chapter 5: Implementation and Testing

This chapter explains the implementation process, technologies used, module descriptions, testing procedures, test cases, and result analysis of the developed system.

Chapter 6: Conclusion and Future Recommendations

This chapter summarizes the overall achievements of the system, project outcomes, limitations, and possible future enhancements for improving the AgroTree Crop Recommendation System.

CHAPTER 2- BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background Study

Agriculture has evolved significantly with the integration of modern technologies, especially in the field of data analysis and machine learning. Traditional agricultural practices mainly depend on farmers' experience and historical knowledge to make decisions regarding crop selection. However, due to increasing climate variability, soil degradation, and unpredictable environmental conditions, such traditional approaches are often insufficient for ensuring optimal productivity.

A Crop Recommendation System is a type of Decision Support System (DSS) that assists farmers in selecting suitable crops based on various parameters such as soil nutrients and climatic conditions. These systems aim to improve agricultural efficiency by providing data-driven insights rather than relying solely on intuition

Machine Learning (ML) Integration:

ML enables systems to identify patterns in environmental and soil data to predict ideal crop choices. Key input parameters include:

Soil Metrics: N, P, K levels and pH.

Climatic Data: Temperature, rainfall, and humidity.

Decision Tree & CART Algorithm:

The system utilizes the CART (Classification and Regression Tree) algorithm. This supervised learning method uses a tree-like structure for classification:

- Binary Splitting: Each node divides data into two branches based on specific conditions.
- Gini Index: This criterion measures node impurity to ensure optimal classification. A lower value indicates a cleaner split, calculated as:

$$Gini(D) = 1 - \sum_{i=1}^c (p_i)^2$$

where p_i is the proportion of samples belonging to class i in the dataset D . In this system, the Gini index serves as the core splitting criterion of the CART Decision Tree. At each node, the feature-threshold pair that yields the lowest weighted Gini impurity across the child nodes is selected as the optimal split, ensuring each division produces the purest possible subsets.

2.2 Literature Review

Several researchers have explored the application of machine learning techniques in agriculture, particularly for crop recommendation and yield prediction. These studies highlight the importance of data-driven approaches in improving agricultural productivity and decision-making.

Bondre and Mahagaonkar developed a system for crop yield prediction and fertilizer recommendation using machine learning algorithms. Their study utilized soil and crop data to analyze patterns and improve prediction accuracy. The results demonstrated that machine learning models can provide more reliable and efficient predictions compared to traditional methods, thereby supporting better agricultural planning and resource utilization [1]

The dataset provided by the Nepal Agricultural Research Council (NARC) includes detailed information on soil properties, crop data, and fertilizer usage. This dataset plays a crucial role in agricultural research by offering real-world data that can be used to train and evaluate machine learning models. It enables researchers to develop systems that are more accurate and relevant to local farming conditions in Nepal [2]

Pradhan discussed site-specific fertilizer recommendations for rice cultivation in Nepal. The study emphasized the importance of analyzing soil conditions at a local level to optimize fertilizer usage and improve crop yield. Although the research does not directly apply machine learning techniques, it highlights the significance of data-driven agricultural practices and supports the development of intelligent decision support systems [3]

Doshi et al. proposed an intelligent crop recommendation system known as AgroConsultant, which uses machine learning algorithms such as Decision Tree, Naive Bayes, and Random Forest. The system analyzes soil and environmental parameters to recommend suitable crops. The study reported that ensemble methods like Random Forest achieved higher accuracy compared to individual models, with overall prediction accuracy reaching above 90% [4]

Bendre et al. explored the use of big data and weather forecasting in precision agriculture. Their study highlighted the role of environmental factors such as temperature, rainfall, and climate variability in influencing crop production. The research suggested that integrating weather data with predictive models can enhance the accuracy of agricultural recommendations and help farmers adapt to changing climatic conditions [5]

From the above studies, it can be observed that machine learning algorithms such as Decision Tree, Random Forest, and Naive Bayes are widely used in crop recommendation systems. Systems that combine soil and climatic data tend to produce more accurate results. However, many existing systems are complex and include multiple features such as fertilizer recommendation and weather prediction, which may reduce usability for farmers.

Therefore, there is a need for a simple and user-friendly system that focuses on essential parameters for crop recommendation. This project addresses this gap by developing a Decision Tree-based crop recommendation system that provides accurate, efficient, and easy-to-use solutions for farmers.

CHAPTER 3-SYSTEM ANALYSIS

3.1 System Analysis

System analysis is the process of studying and understanding the requirements, functionality, and feasibility of the proposed system. It helps in identifying what the system should do and how it should perform. For the AgroTree Crop Recommendation System, system analysis focuses on understanding user needs, system requirements, and overall system behavior.

3.1.1 Requirement Analysis

Requirement analysis defines the functionalities and constraints of the system. It is divided into functional and non-functional requirements.

3.1.1.1 Functional Requirements

The functional requirements define the core operations and services provided by the AgroTree Crop Recommendation System. These requirements describe how the system interacts with users and performs crop prediction, data management, and authentication processes.

The functional requirements of the AgroTree Crop Recommendation System are as follows:

- User registration and login authentication
- Secure session management using Supabase
- Input of soil and climatic parameters such as Nitrogen (N), Phosphorus (P), Potassium (K), pH, temperature, humidity, and rainfall
- Validation of user input values within acceptable ranges
- Crop recommendation using the CART-based Decision Tree algorithm
- Display of predicted crop results with confidence score
- Storage of prediction history in the database
- Viewing previous prediction records by users

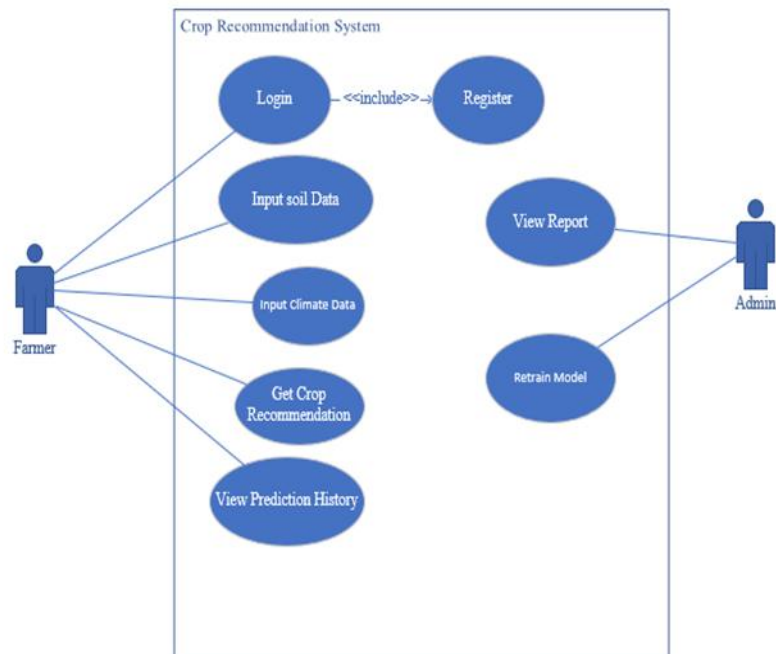


Figure 3.3: Use Case Diagram of AgroTree

Table 3.2: Descriptive Use Case of Register

Use Case	Register
Primary Actor	Farmer
Secondary Actor	
Precondition	Farmer is not already registered
Postcondition	1. New farmer account is created successfully. 2. Farmer can log in to the system.
Failure Condition	- Email/username already exists. - Invalid or incomplete input data.

Table 3.3: Descriptive Use Case of Login

Use Case	Login
Primary Actor	Farmer / Admin
Secondary Actor	
Precondition	User must be registered.
Postcondition	User is successfully logged into the dashboard.
Failure Condition	• Incorrect username or password. • Account not registered.

Table 3.4: Descriptive Use Case of Input Soil Data

Use Case	Input Soil Data
Primary Actor	Farmer
Secondary Actor	
Pre Condition	Farmer must be logged in.
Post Condition	Soil data is stored in the database.
Failure Condition	• Missing required soil parameters. • Invalid numeric values. • Database saving error.

Table 3.5: Descriptive Use Case of Input Climate Data

Use Case	Input Climate Data
Primary Actor	Farmer
Secondary Actor	System
Pre Condition	Farmer must be logged in.
Post Condition	Climate data is successfully stored.
Failure Condition	• Incomplete climate information. • Invalid data format.

Table 3.6: Descriptive Use Case of Crop Recommendation

Use Case	Get Crop Recommendation
Primary Actor	Farmer
Secondary Actor	System
Pre Condition	• Farmer must be logged in. • Soil and climate data must be available.
Post Condition	• Suitable crop recommendation is displayed. • Prediction is stored in history.
Failure Condition	• Required data not available. • ML model not loaded or trained. • System processing error.

Table 3.7: Descriptive Use Case of view Prediction History

Use Case	View Prediction History
Primary Actor	Farmer
Secondary Actor	System
Pre Condition	Farmer must be logged in
Post Condition	Previous prediction records are displayed.
Failure Condition	• No prediction history found. • Database retrieval error.

3.1.1.2 Non-Functional Requirements

The non-functional requirements of the AgroTree Crop Recommendation System are as follows:

- Fast response time for generating crop predictions
- Efficient page loading and smooth system performance

- Support for multiple users simultaneously
- Secure data transmission using HTTPS
- Secure password management through Supabase authentication
- Privacy and protection of user prediction history

3.1.2 Feasibility Analysis

Feasibility analysis evaluates whether the AgroTree Crop Recommendation System is practical, achievable, and suitable for implementation within available resources, technologies, and time constraints.

i. Technical Feasibility

The AgroTree system is technically feasible because it is developed using widely available and reliable technologies such as Python, Streamlit, and Scikit-learn. The machine learning model is implemented using the Decision Tree algorithm based on the CART approach, which is suitable for classification problems involving structured agricultural datasets. The system uses a Kaggle agricultural dataset containing soil nutrients and climatic parameters required for crop prediction. The dataset is compatible with machine learning preprocessing and model training techniques. Streamlit provides a lightweight framework for developing interactive web applications, while Supabase is used for secure user authentication and prediction history storage. The hardware and software requirements of the system are minimal and can be supported on standard computers with internet connectivity. Therefore, the technologies, tools, and development environment required for implementing the system are readily available and technically achievable.

ii. Operational Feasibility

The AgroTree system is operationally feasible because it is designed with simplicity and usability in mind. Users can easily enter soil and climatic parameters through a user-friendly web interface and receive real-time crop recommendations without requiring advanced technical knowledge.

The system includes features such as secure login, prediction history, and clear result display, which improve usability and operational efficiency. Error messages and validation mechanisms help users provide valid input values during prediction.

To further support users, the system can include:

FAQ (Frequently Asked Questions): Helps users understand how to use the system, enter parameters, and interpret prediction results.

SBD (System Behavior Description): Describes how the system processes user inputs, performs predictions, and generates outputs, improving transparency and user understanding.

Since the system automates crop recommendation based on data-driven analysis, it reduces dependency on traditional guesswork and improves agricultural decision-making efficiency

iii. Economic Feasibility

The AgroTree system is economically feasible because it is developed primarily using open-source technologies and free development platforms. Tools such as Python, Streamlit, and Scikit-learn are freely available and do not require licensing costs. Additionally, Supabase free-tier services are used for authentication and database management, reducing infrastructure expenses. The development cost mainly consists of development effort, internet usage, report preparation, and miscellaneous academic expenses. No expensive hardware or specialized infrastructure is required for system implementation or deployment.

Table 3.8: Estimated Development Cost

Cost Item	Estimated Cost (NPR)
Development effort (3 members × 200 hrs × NPR 100/hr)	NPR 60,000
Printing and report preparation	NPR 1,500
Internet connectivity (2.5 months)	NPR 2,000
Miscellaneous expenses (binding, stationery, travel)	NPR 1,000
Total Estimated Cost	NPR 64,500

The use of free and open-source technologies significantly reduces the overall development and operational cost, making the system economically practical.

iv. Schedule Feasibility

The AgroTree system is schedule feasible because the project development tasks were planned and managed using an Agile development approach. The project activities were divided into multiple stages including requirement analysis, data collection, preprocessing, model development, system implementation, testing, and documentation. The Decision Tree algorithm was selected because of its relatively simple implementation and faster training process compared to more complex machine

learning models. Streamlit also accelerated frontend development due to its rapid prototyping capabilities. The project tasks were completed incrementally within the allocated academic schedule through proper planning, teamwork, and continuous testing. Therefore, the system development was achievable within the available project timeline.

3.1.3 Analysis

The system follows an object-oriented approach, where different components are designed as independent modules.

Class Diagram

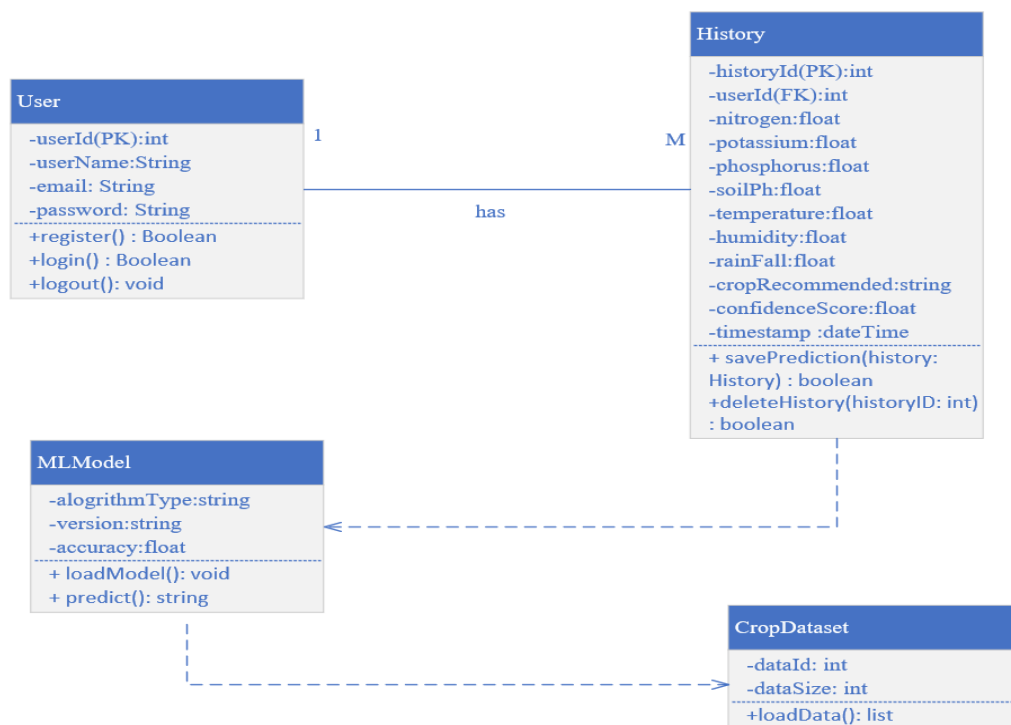


Figure 3.4: Class Diagram of AgroTree

This class diagram serves as the blueprint for a Crop Recommendation System, mapping out how the system handles user sessions, stores prediction history, and utilizes machine learning for decision-making.

Activity Diagram

This Activity Diagram illustrates the complete workflow of the machine learning model development and deployment process for the Crop Recommendation System. It demonstrates how raw agricultural data is transformed into a trained model that is

integrated into the application for making crop predictions.

Process Flow

The process starts with Agriculture Dataset Collection, followed by Data Preprocessing (cleaning, handling missing values, IQR outlier treatment, and log transformation). The dataset is then split into 70% training, 10% validation, and 20% testing. The CART model is trained and evaluated. If the model accuracy is not acceptable, the process returns to retrain the model. Otherwise, the trained model is saved and loaded into the Streamlit UI for crop prediction, completing the workflow.

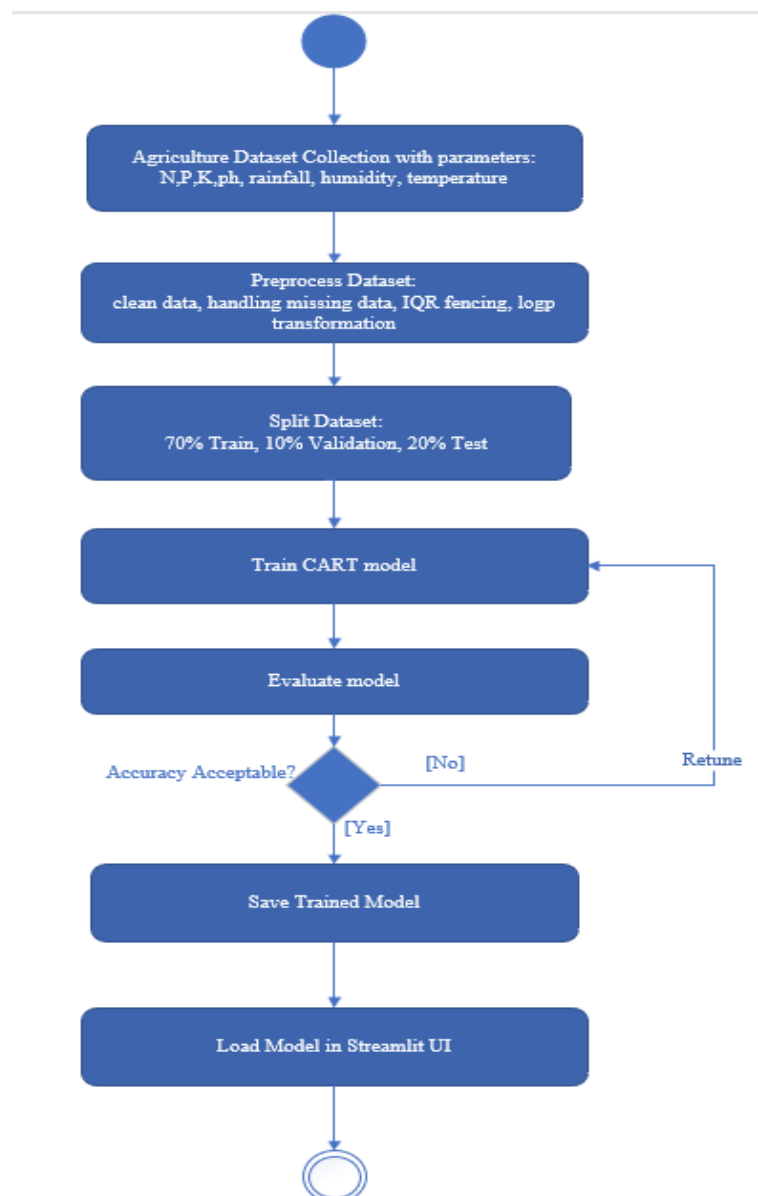


Figure 3.5:Activity Diagram

Sequence Diagram

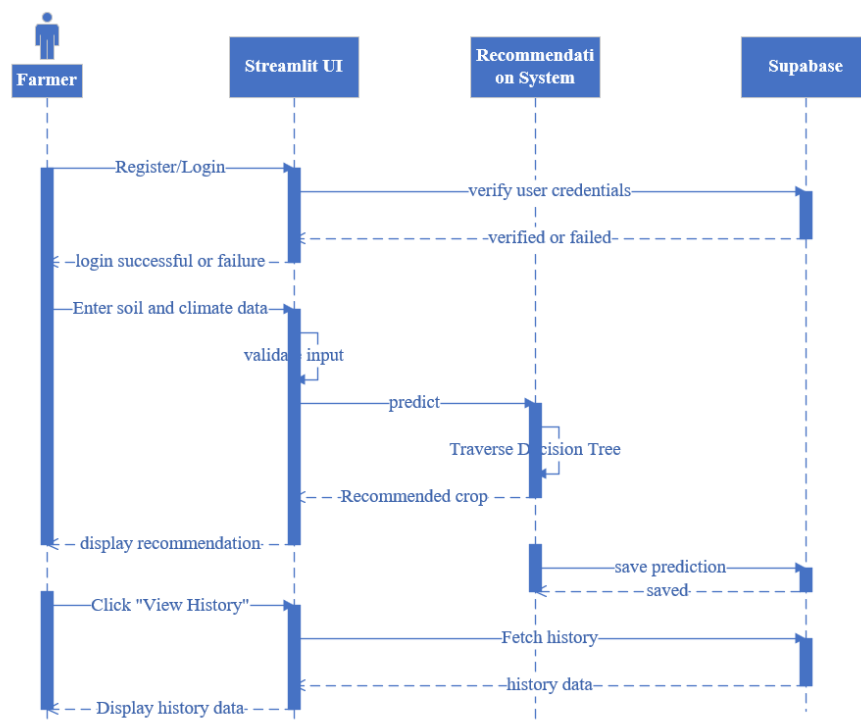


Figure 3.6: Sequence Diagram of AgrooTree

This Sequence Diagram outlines the end-to-end interaction flow of the Crop Recommendation System from the farmer's perspective. It begins with the Farmer authenticating through the Streamlit UI, which verifies credentials against the Database. Once logged in, the farmer inputs environmental and soil parameters, which are validated by the UI before being sent to the Recommendation System. The system processes this input by traversing the Decision Tree to generate a crop recommendation, which is then displayed to the farmer and automatically saved to the Database. Finally, the diagram illustrates the user retrieving their past activity by triggering the "View History" function, which fetches and displays the stored records from the database back to the user.

Persistence Class Diagram

This Persistence Class Diagram outlines the database structure and data storage mechanism for the system. It visualizes how user identities and machine learning outputs are permanently stored and linked within the Supabase backend.

Database Structure:

Supabase Auth User Table: Handles user authentication by storing secure login credentials.

PredictionHistory Table: Records the complete transaction, storing both the

environmental inputs (soil/climate data) and the resulting crop recommendation.

Profiles Table: Store user information such as full name, email and role.

Relationship:

Supabase Auth $\square$ Profile: One - to - One

Profiles $\square$ PredictionHistory: One – to - Many

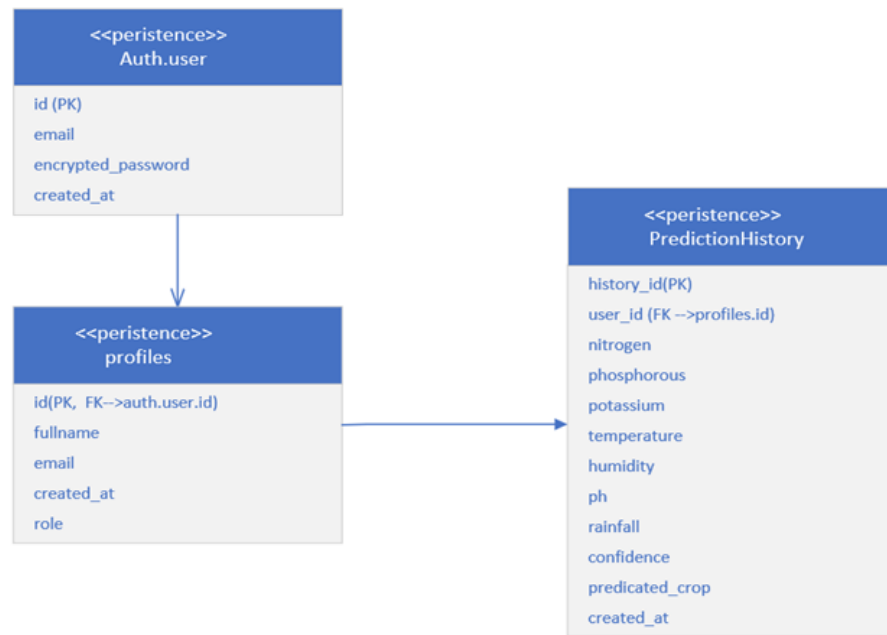


Figure 3.7: Persistence Class Diagram of AgroTree

CHAPTER 4- SYSTEM DESIGN

4.1 Design

The AgroTree Crop Recommendation System follows an object-oriented design approach. The system design phase focuses on refining the analysis models and defining the overall structure, components, interactions, and deployment of the system. Various UML diagrams are used to represent the system architecture, workflows, object interactions, and deployment environment.

Component Diagram

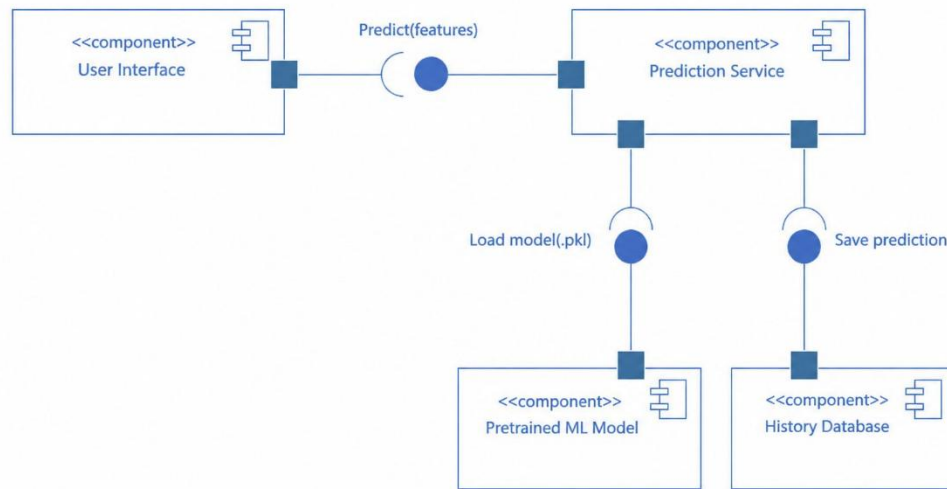


Figure 4.8: Component Class Diagram of AgroTree

This component diagram illustrates the high-level architecture of the Crop Recommendation System, showcasing how four key modules—User Interface, Prediction Model, Pretrained ML Model, and History Database—interact to deliver results. It utilizes UML notation, specifically Provided (PI) and Required (RI) interfaces, to define the communication channels between these components. This design ensures a clean separation of concerns, where the user interface handles data input, the prediction model serves as the central logic controller, the ML model provides the analytical engine, and the database manages persistent storage, resulting in a modular and maintainable system.

Deployment Diagram

This deployment diagram illustrates the physical infrastructure and software distribution of the Crop Recommendation System across three distinct environments. It shows the Farmer Device (Client) running a web browser that communicates via HTTP with the Web/Application Server, which hosts the Streamlit application and the pre-

trained Decision Tree model (.pkl). Simultaneously, the Application Server interacts with a separate Database Server using read/write operations to manage data within a PostgreSQL database. By mapping these artifacts to their respective hardware environments, the diagram demonstrates how the system architecture supports remote client access, centralized processing, and persistent data storage.

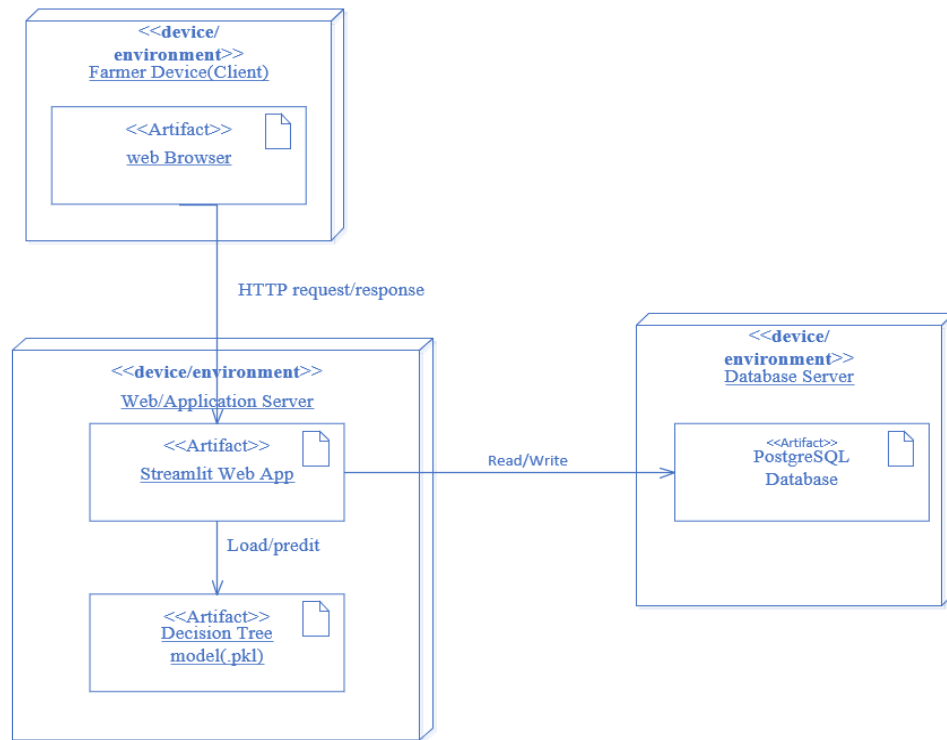


Figure 4.9: Deployment Diagram of AgroTree

4.2 Algorithm Details

The AgroTree Crop Recommendation System uses the Decision Tree algorithm based on the CART (Classification and Regression Tree) methodology for predicting the most suitable crop according to soil nutrients and climatic conditions. The algorithm was selected because of its high interpretability, faster prediction capability, low computational complexity, and effectiveness in handling structured agricultural datasets. The Decision Tree model classifies crops using agricultural parameters such as Nitrogen (N), Phosphorus (P), Potassium (K), pH value, temperature, humidity, and rainfall. The algorithm learns hidden relationships between these environmental factors and crop suitability from historical agricultural data.

4.2.1 Mathematical Model of the System

The AgroTree system can be mathematically represented as a classification function:

$$f(X) = Y$$

Where:

X = Input feature vector

Y = Predicted crop class

The input vector consists of seven agricultural parameters:

$$X = \{x_1, x_2, x_3, x_4, x_5, x_6, x_7\}$$

Where:

Table 4.9: Input Feature Variables of the AgroTree System

Variable	Description
x_1	Nitrogen (N)
x_2	Phosphorus (P)
x_3	Potassium (K)
x_4	pH Value
x_5	Temperature
x_6	Humidity
x_7	Rainfall

The output variable is:

$$Y = \{CropType\}$$

The model predicts the most suitable crop by recursively partitioning the feature space into decision regions.

4.2.2 Decision Tree Architecture

The Decision Tree algorithm organizes decisions in a hierarchical tree structure.

The architecture consists of:

1. Root Node

The root node represents the complete agricultural dataset used for training. At this stage, the algorithm evaluates all available features and determines the best feature for the initial split.

2. Internal Nodes

Internal nodes represent decision conditions based on feature thresholds. Each node divides the dataset into smaller subsets according to the selected condition.

Each split attempts to increase the purity of resulting subsets.

3. Leaf Nodes

Leaf nodes represent the final prediction output of the model. Each leaf node corresponds to a crop class such as:

- Rice
- Wheat
- Maize
- Cotton
- Coffee

The final predicted crop is determined based on the majority class present in the leaf node.

4.2.3 CART (Classification and Regression Tree)

The AgroTree system implements the Classification and Regression Tree (CART) algorithm from scratch for multi-class crop classification. The CART algorithm recursively partitions the training dataset using binary decision rules.

The main characteristics of the implemented CART algorithm are:

- Binary splitting
- Recursive partitioning
- Gini impurity for measuring node impurity
- Greedy selection of the best feature and threshold
- Majority-class prediction at leaf nodes

Since AgroTree performs crop classification, the classification component of CART is used.

4.2.4 Binary Splitting Process

At each node, the algorithm searches for:

Best Feature: The feature that produces the lowest weighted Gini impurity when all available features are evaluated.

Best Threshold: The threshold value that creates the split with the lowest weighted Gini impurity for the selected feature.

Suppose the feature is:

$$pH \leq 6.5$$

Then:

- Left branch $\rightarrow$ values satisfying the condition

- Right branch → values not satisfying the condition

The process continues recursively until stopping conditions are reached.

4.2.5 Gini Index Calculation

The AgroTree system uses the Gini Index to measure node impurity.

The Gini Index formula is:

$$Gini(D) = 1 - \sum_{i=1}^c (p_i)^2$$

Where:

Table 4.10: Symbols Used in Gini Index Formula

Symbol	Meaning
D	Dataset at current node
c	Number of crop classes
p_i	Probability of class i

Table 4.11: Interpretation of Gini index Values

Gini Value	Interpretation
0	Completely pure node
Close to 1	Highly impure node

The algorithm selects the split with the minimum weighted Gini impurity.

4.2.6 Weighted Gini Calculation

For every possible split:

$$Gini_{split} = \frac{n}{n} Gini(+) + \frac{n}{n} Gini(-)$$

Where:

Table 4.12: Symbols Used in Weighted Gini Calculation

Symbol	Meaning
n	Total samples
n	Samples in left node
n_{right}	Samples in right node

The split producing the smallest weighted impurity is selected.

4.2.7 Recursive Partitioning

After selecting the best split:

1. Dataset is divided into subsets
2. Each subset becomes a child node
3. The same splitting process repeats recursively

This recursive partitioning continues until stopping criteria are met.

4.2.8 Stopping Conditions

The recursive tree-building process stops when one or more of the following conditions are satisfied:

- The maximum tree depth of 15 is reached.
- The number of samples in a node is less than the required minimum for splitting.
- A node contains samples from only one crop class.
- The minimum leaf-size constraint prevents a valid split.
- No valid feature-threshold split can improve the node impurity.

When the recursive splitting process stops, the node becomes a leaf node. The majority crop class within that node is then used as the final prediction.

The final prediction process can be represented as:

$$\hat{y} = f(x)$$

where the input feature vector travels from the root node through a sequence of binary decision conditions until it reaches a leaf node containing the predicted crop class.

4.2.9 Feature Importance

The algorithm automatically identifies the most influential agricultural parameters during training.

Features contributing to higher impurity reduction obtain greater importance.

Common important features in crop prediction include:

- Rainfall
- Temperature
- Nitrogen
- pH value

Feature importance helps improve model interpretability.

4.2.9 Heuristic Penalty

Even after Platt scaling, extremely rare or borderline values can produce overconfident predictions. A simple rule-based penalty reduces the displayed confidence for such inputs:

- **Near-boundary penalty (5% per feature)** – If a value lies within 8% of the

global minimum or maximum (from VALID_RANGES), a 5% penalty is applied.

- **Unusual range penalty (3% per feature)** – If a value is below the “unusually low” threshold or above the “unusually high” threshold (derived from per-crop means in the dataset), a 3% penalty is applied.

The total penalty is capped at 40%. The final confidence is:

$$\text{final confidence} = \text{calibrated confidence} \times (1 - \text{total penalty})$$

This penalty does not change the predicted crop; it only lowers the confidence and triggers a warning message to the user, explaining why the confidence is reduced.

4.2.10 Training Phase

The AgroTree model training process consists of the following steps:

Step 1: Dataset Preparation

The agricultural dataset is cleaned and preprocessed. The input features used are:

[N, P, K, Temperature, Humidity, pH, Rainfall]

Step 2: Dataset Splitting

The dataset is divided into three subsets:

70% Training Data – Used to train the CART Decision Tree.

20% Validation Data – Used to evaluate and tune the model during development.

10% Testing Data – Used for final performance evaluation.

Step 3: Model Training

The CART algorithm uses the training data to:

- Calculate Gini impurity.
- Select the best feature and threshold.
- Create binary decision nodes.
- Recursively construct the Decision Tree.
- Assign crop classes to leaf nodes.

The validation set is used to evaluate the model during development, while the test set is reserved for the final evaluation.

4.2.11 Prediction Phase

When a user enters agricultural input values, the prediction process follows these steps:

1. Input Validation – Checks whether the input values are valid.
 2. Preprocessing – Applies the same preprocessing used during training.
 3. Feature Vector Creation – Converts the inputs into:

$$f(x) = [N, P, K, \text{Temperature}, \text{Humidity}, \text{pH}, \text{Rainfall}]$$
 4. Tree Traversal – The input follows binary decision paths through the trained CART tree.
 5. Crop Prediction – The final leaf node provides the predicted crop.
- The prediction function is represented as:

$$\hat{y} = f(x)$$

where $f(x)$ is the input feature vector and $\hat{y}$ is the predicted crop label.

Example:

$$f(x) = [90, 42, 43, 25.5, 80, 6.5, 200]$$

4.2.12 System Architecture of Prediction Process

The prediction architecture of AgroTree follows the following workflow:

1. User enters agricultural parameters
2. Streamlit frontend sends input to backend
3. Input validation module checks ranges
4. Decision Tree model processes data
5. Prediction result is generated
6. Result is displayed to the user
7. Prediction history is stored in Supabase database

CHAPTER 5- IMPLEMENTATION AND TESTING

5.1 Implementation

The implementation phase focuses on the development of the AgroTree Crop Recommendation System using machine learning and web technologies. The system was implemented as a web-based application that allows users to enter soil and climatic parameters and receive crop recommendations in real time. The implementation process included dataset preprocessing, model training, frontend development, backend integration, authentication management, database connectivity, and deployment preparation. The system follows a modular architecture where each module performs a specific task such as user authentication, prediction processing, data storage, and dataset management.

5.1.1 Tools Used

The following tools and technologies were used for the implementation:

- **Programming Language:** Python
- **Frontend Framework:** Streamlit
- **Machine Learning:** Custom CART implementation (from scratch) for final model; Scikit-learn used only for hyperparameter tuning (K-fold validation)
- **Data Processing:** Pandas, NumPy
- **Database & Authentication:** Supabase
- **Version Control:** Git / GitHub
- **Development Environment:** Jupyter Notebook (exploration), VS Code
- **Model Storage:** Pickle (.pkl) files – model, scaler, medians, fences

5.1.2 Implementation Details of Modules

The system is divided into several modules for better maintainability and scalability.

1. User Authentication Module

The authentication module manages user registration, login, session management, and logout functionality using Supabase authentication services.

Functions Performed

- User registration using email and password
- Secure login authentication
- Session token management
- Logout handling

Implementation Details

Supabase authentication APIs were integrated into the Streamlit application to securely manage user accounts and sessions. Password hashing and authentication security are handled automatically by Supabase.

Input

- Email
- Password

Output

- Authenticated user session

2. Crop Recommendation Module

This is the core module of the AgroTree system responsible for predicting suitable crops.

Functions Performed

- Accept user input parameters
- Validate input ranges
- Load trained Decision Tree model
- Generate crop predictions
- Display recommendation result

Input Parameters

- Nitrogen (N)
- Phosphorus (P)
- Potassium (K)
- pH value
- Temperature
- Humidity
- Rainfall

Implementation Details

The CART-based Decision Tree model was trained using the agricultural dataset and serialized using Joblib. During prediction, the saved model file is loaded into the Streamlit application and used for real-time classification.

The prediction workflow includes:

1. Input collection
2. Validation
3. Feature vector creation

4. Model prediction
5. Result display

3. Dataset Management Module (Admin)

The dataset management module allows administrators to manage agricultural records.

Functions Performed

- View dataset
- Add new records
- Edit existing records
- Delete records

Implementation Details

The dataset management functionality was implemented using database operations integrated with Supabase. Input validation mechanisms ensure data consistency and integrity.

4. Prediction History Module

This module stores and displays previous prediction records for authenticated users.

Functions Performed

- Store prediction history
- Retrieve user-specific records
- Display previous predictions

Implementation Details

Each prediction generated by the system is stored in the database with:

- Timestamp
- User ID
- Input parameters
- Predicted crop

Supabase database services are used for storing and retrieving prediction history.

5. Model Training Module

This module handles machine learning model training and evaluation.

Implementation Steps

1. Dataset loading
2. Data preprocessing
3. Feature selection
4. Train-test splitting
5. Decision Tree model training

6. Accuracy evaluation
7. Model serialization

Algorithm Used

- CART (Classification and Regression Tree)

6. Preprocessing Module

Defined in `src/preprocess.py`. This module loads raw data, applies IQR winsorisation per crop, imputes missing values with per-crop medians, applies `log1p` transform to K and rainfall, performs standard scaling, and splits the data into 70% training, 10% validation, and 20% test sets. It saves the cleaned dataset, scaler, medians, and fences for reuse.

7. Heuristic Penalty Module

Implemented inside `src/predict.py`. It uses the ranges from `src/validations.py` to calculate near-boundary and unusual-value penalties and applies them to the calibrated confidence.

5.2 Testing

Testing was performed to verify the correctness, reliability, performance, and usability of the AgroTree system. Both unit testing and system testing were conducted to ensure that each module functions properly and that the integrated system performs as expected.

5.2.1 Test Cases for Unit Testing

Unit testing in the AgroTree system was performed to verify that individual modules and functions worked correctly in isolation. The major components tested included the CART model loading process, prediction functionality, prediction output type validation, and exception handling for model loading failures. In addition, authentication modules such as user registration and login were tested using different valid and invalid scenarios to ensure reliability and robustness.

Table 5.13: Test case for User Registration

TC ID	Scenario	Steps	Input Data	Expected Result	Actual Result	P/F
TC01	Register with valid credentials	1. Open app 2. Click Register 3. Enter details 4. Submit	Full Name:dixan kc Email: dixankccc@gmail.com Password: Agro@2026	Account created and redirected to login page	User account created successfully and login page displayed	Pass
TC02	Register with weak password	1. Open app 2. Click Register 3. Enter details 4. Submit	Full Name:Prajwal Pathak Email: prajwal@gmail.com Password: 123	Error message displayed for weak password	Validation message shown: "Password must be at least 8 characters"	Pass
TC03	Register with already used email	1. Open app 2. Click Register 3. Use existing email	Full Name:Chetan Bhattarai Email: chetanbhattarai23@gmail.com Password: Agro@2026	Error message indicating email already registered	Error displayed: "User may already registered"	Pass

Table 5.14: Test case for User login

TC ID	Scenario	Steps	Input Data	Expected Result	Actual Result	P/F
TC01	Login with valid credentials	1. Open app 2. Enter email & password 3. Click Login	Email: dixankcc@gmail.com Password: rightP@ss1	User authenticated and redirected to dashboard	Recommendation dashboard opened successfully	Pass
TC02	Login with wrong password	1. Open app 2. Enter email 3. Enter wrong password	Email: dixankccc@gmail.com Password: wrongp@ss	Error message: Invalid email or password	Authentication failed message displayed	Pass
TC03	Login with unregistered email	1. Open app 2. Enter unregistered email	Email: unknown@abc.com Password: any123	Error message: Invalid email or password	Login denied with invalid credential message	Pass

Table 5.15: Test Case for Crop Recommendation Model

TC ID	Scenario	Test Description	Expected Result	Actual Result	P/F
TC01	Model loads successfully	Verify that the trained model file loads correctly	Model loads without error	Model loaded successfully from saved .pkl file	Pass
TC02	Model has predict method	Verify that the loaded model exposes predict() functionality	Predict method available	predict() method detected in loaded model	Pass
TC03	Model accepts	Provide sample agricultural input	Prediction generated	Model returned crop prediction	Pass

	valid input	data to the model	successfully	successfully	
TC04	Prediction output type validation	Verify that prediction output is returned as crop name string	Output type is string	Output received as string value (e.g., “rice”)	Pass
TC05	Model load failure handling	Simulate missing/corrupted model file	FileNotFoundError raised correctly	FileNotFoundError exception triggered successfully	Pass

Fig 5.1: Result of Unit Testing for Crop Recommendation model

```

PS C:\Users\Dikshan\Desktop\FinalProject\Crop-Recommendation-System> python -m pytest test.py -v
===== test session starts =====
platform win32 -- Python 3.13.2, pytest-8.4.2, pluggy-1.6.0 -- C:\Users\Dikshan\AppData\Local\Programs\Python\Python313\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\Dikshan\Desktop\FinalProject\Crop-Recommendation-System
plugins: anyio-4.12.1, langsmith-0.7.29
collected 5 items

test.py::test_TC01_model_loads PASSED [ 20%]
test.py::test_TC02_model_has_predict PASSED [ 40%]
test.py::test_TC03_model_accepts_input PASSED [ 60%]
test.py::test_TC04_model_output_type PASSED [ 80%]
test.py::test_TC05_model_load_failure PASSED [100%]

===== 5 passed in 0.91s =====
PS C:\Users\Dikshan\Desktop\FinalProject\Crop-Recommendation-System>

```

Table 5.16: Test case for attribute in Crop Recommendation Form

TC ID	Scenario	Steps	Input Data	Expected Result	Actual Result	P/F
TC01	Valid crop prediction with balanced soil values	1. Login 2. Enter all parameters 3. Click Get Recommendation	N:85 P:50 K:40 Temp:22°C Hum:82% pH:7.0 Rain:230mm	System should generate suitable crop recommendation	System displayed crop recommendation successfully with confidence score	Pass

TC02	N value out of range	1. Login 2. Enter N=200 3. Click Get Recommendation	N:200 P:50 K:40 Temp:22°C Hum:82% pH:7.0 Rain:230mm	Error: “N must be between 0 and 155 mg/kg”	Validation error displayed: “Nitrogen value must be between 0 and 155 mg/kg”	Pass
TC03	pH out of acceptable range	1. Login 2. Enter pH=12.5 3. Click Get Recommendation	N:85 P:50 K:40 Temp:22°C Hum:82% pH:10.5 Rain:230mm	Error: “pH must be between 2.9 and 11”	Validation error displayed: “Soil pH must be between 2.9 and 11”	Pass

5.2.2 Test Cases for System Testing

System testing was performed to evaluate the complete AgroTree application as an integrated system. The testing verified that all modules, including user authentication, crop recommendation prediction, input validation, prediction history, and database connectivity, functioned correctly together. Different test scenarios were executed to ensure the system produced accurate outputs, handled errors properly, and maintained stable performance during real-time usage.

Table 5.17: Test case for whole system Testing

TC ID	Scenario	Steps	Input Data	Expected Result	Actual Result	P/F
ST01	Complete user workflow	1. Register account 2. Login 3. Enter crop parameters 4. Click Get Recommendation	Valid user credentials and crop data	System should complete full workflow successfully	User registered, logged in, crop predicted, and redirected successfully	Pass

ST02	Automatic prediction history saving	1. Login 2. Enter crop parameters 3. Click Get Recommendation	Valid crop recommendation input	Prediction should automatically save to history database	Prediction saved automatically and displayed in history section	Pass
ST03	Invalid login attempt	1. Open login page 2. Enter wrong password 3. Click Login	Email: dikshan@gmail.com Password: wrongpass	System should deny access and display error	Authentication failed message displayed successfully	Pass
ST04	Invalid crop input handling	1. Login 2. Enter invalid parameter values 3. Click Get Recommendation	N=200, pH=12.5	System should validate input and prevent prediction	Validation error messages displayed correctly	Pass
ST05	Database connectivity test	1. Login 2. Generate prediction 3. Open history tab	Valid crop prediction data	System should connect with Supabase database successfully	Prediction history retrieved successfully from database	Pass
ST06	Session management	1. Login 2. Refresh application 3. Access dashboard	Valid logged-in user session	User session should remain active	Dashboard remained accessible without re-login	Pass
ST07	Logout functionality	1. Login 2. Click Logout	Valid authenticated session	User should be logged out and redirected	Session terminated and login page displayed successfully	Pass

5.3 Result Analysis

The AgroTree Crop Recommendation System was successfully implemented and evaluated as a machine learning-based web application for crop prediction using soil nutrients and climatic parameters. The custom CART-based Decision Tree model achieved a test accuracy of 94.3%, with weighted average precision of 94.4%, recall of 94.3%, and F1-score of 94.3%. These results demonstrate strong and balanced predictive performance across the 22 crop classes.

The model performed particularly well for muskmelon (F1-score: 0.991), watermelon (0.989), and chickpea (0.989). Although the dataset exhibited a moderate class imbalance with a ratio of 2.56, no significantly low-performing crop classes were identified, indicating good generalization capability. The preprocessing pipeline, consisting of IQR winsorisation, median imputation, and log transformations, contributed to improved data quality and model stability.

The system was further integrated with a Streamlit-based interface and Supabase for secure user authentication and prediction history management. Overall, AgroTree successfully met its objectives by providing an accessible, data-driven crop recommendation system that can support informed agricultural decisions, optimize resource utilization, reduce crop selection risks, and promote sustainable precision agriculture.

```
Accuracy: 0.9430
Macro Precision: 0.9361
Macro Recall: 0.9311
Macro F1-score: 0.9328
Weighted Precision: 0.9437
Weighted Recall: 0.9430
Weighted F1-score: 0.9427
Low-performance classes: None
2026-07-23 15:33:44,703 INFO agrotree: Test Accuracy: 0.9430
2026-07-23 15:33:44,703 INFO agrotree: Macro Precision: 0.9361
2026-07-23 15:33:44,703 INFO agrotree: Macro Recall: 0.9311
2026-07-23 15:33:44,703 INFO agrotree: Macro F1-score: 0.9328
2026-07-23 15:33:44,703 INFO agrotree: Weighted Precision: 0.9437
2026-07-23 15:33:44,703 INFO agrotree: Weighted Recall: 0.9430
2026-07-23 15:33:44,703 INFO agrotree: Weighted F1-score: 0.9427
2026-07-23 15:33:44,703 INFO agrotree: Low-performance classes: None
```

```
Cross-validation results:
Fold 1 accuracy: 0.9474
Fold 2 accuracy: 0.9466
Fold 3 accuracy: 0.9493
Fold 4 accuracy: 0.9439
Fold 5 accuracy: 0.9456
Mean accuracy: 0.9466
Standard deviation: 0.0018
```

CHAPTER 6-CONCLUSION AND FUTURE RECOMMENDATION

6.1 Conclusion

The AgroTree Crop Recommendation System was successfully developed as a machine learning-powered web application for intelligent crop prediction based on soil nutrients and climatic parameters. The CART-based Decision Tree model, trained on a Kaggle dataset, achieved a test accuracy of 94.3%, with weighted average precision of 94.4%, recall of 94.3%, and F1-score of 94.3%. The robust preprocessing pipeline including IQR winsorisation, median imputation, and log transformations ensured data quality and model stability.

The model demonstrated strong performance across all 22 crop classes, with muskmelon (F1: 0.991), watermelon (F1: 0.989), and chickpea (F1: 0.989) among the top performers. While the dataset showed a moderate imbalance (ratio 2.56), no low-performance classes were identified, confirming the model's generalization capability. The Streamlit-based interface, integrated with Supabase for authentication and history management, provides farmers with an accessible, data-driven decision-support tool. By replacing traditional guesswork with scientific recommendations, AgroTree helps optimize resource utilization, reduce crop failure risk, and promote sustainable agricultural practices successfully demonstrating the practical value of machine learning in precision agriculture.

6.2 Future Recommendations

Although the system performs effectively, there are several areas where it can be further improved and extended:

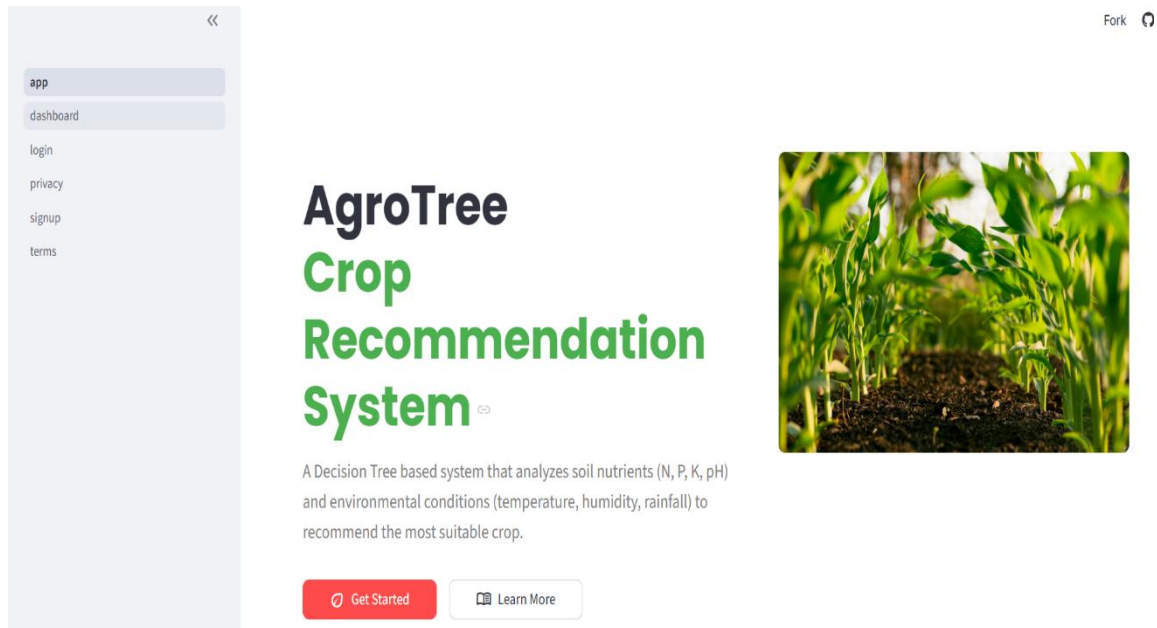
- **Integration of Real-Time Weather Data:**
Incorporating live weather APIs can improve prediction accuracy by considering real-time environmental conditions.
- **Use of Advanced Machine Learning Models:**
Implementing advanced algorithms such as Random Forest or Gradient Boosting can enhance model performance and accuracy.
- **Mobile Application Development:**
Developing a mobile version of the system would increase accessibility for farmers in rural areas.

- **Multilingual Support:**
Adding support for multiple languages (such as Nepali) can make the system more user-friendly and accessible.
- **Fertilizer Recommendation System:**
Extending the system to include fertilizer suggestions based on soil nutrients can provide a more comprehensive solution.
- **Scalability and Deployment:**
Deploying the system on cloud platforms can allow it to handle a larger number of users efficiently.
- **Dataset Expansion:**
Using larger and more diverse datasets can improve model generalization and accuracy.

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APPENDIX



Welcome Back

Log in to your AgroTree account

Email Address

Password



Press Enter to submit form

Log In

Don't have an account?

 Sign Up Instead

Crop Recommendation Dashboard

Make Prediction History (14)

Enter the soil and environmental parameters below to get a crop recommendation.

Soil Nutrients

Nitrogen (N) • 1 – 155 mg/kg

50.00 - +

Phosphorus (P) • 10 – 145 mg/kg

50.00 - +

Potassium (K) • 10 – 240 mg/kg

50.00 - +

Climate Conditions

Temperature • 8 – 45 °C

25.00 - +

Humidity • 14 – 100 %

60.00 - +

Soil Chemistry & Water

Soil pH • 3.5 – 9.5

6.50 - +

Rainfall • 20 – 850 mm

100.00 - +

Get Recommendation

Prediction Complete

The most suitable crop for these conditions is: JUTE

Confidence: 57.1%

Prediction saved to history!

Crop Recommendation Dashboard

Make Prediction History (14)

Total Predictions 15	Unique Crops 6	Most Predicted Pigeonpeas
Search crop... Rice, Wheat, etc		Sort Newest
> Jute — May 13, 2026 at 21:46		
> Pigeonpeas — May 03, 2026 at 11:56		
> Pigeonpeas — May 03, 2026 at 11:56		
> Pigeonpeas — May 02, 2026 at 10:09		
> Pigeonpeas — Apr 08, 2026 at 09:03		